EXPERIMENT : 3

3.a Data visualization using line plot.

AIM:

Conduct an experiment to show data visualization using line plot.

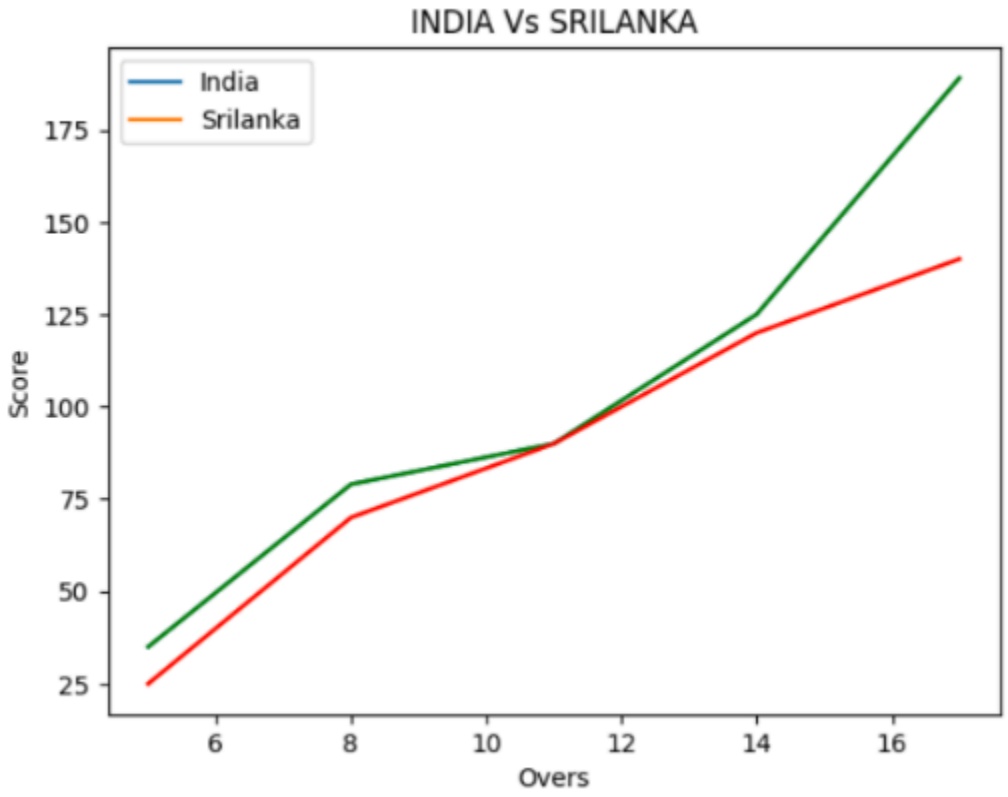
DESCRIPTION:

Take any sample data either through csv file or data fetched directly through code.

CODE:

```
import matplotlib.pyplot as plt
Overs=list(range(5,20,3))
Team_A=[35,79,90,125,189]
Team_B=[25,70,90,120,140]
plt.plot(Overs,Team_A)
plt.plot(Overs,Team_B)
plt.title("INDIA Vs SRILANKA")
plt.xlabel("Overs")
plt.ylabel("Score")
plt.legend(["India","Srilanka"])
plt.plot(Overs,Team_A,color="green",label="INDIA")
plt.plot(Overs,Team_B,color="red",label="SRILANKA")
plt.show()
```

OUTPUT:



RESULT:

The data visualization of India vs Srilanka was successfully performed using line plot.

3.b Data visualization using bar chart

AIM:

Conduct an experiment to show data visualization using bar chart

.

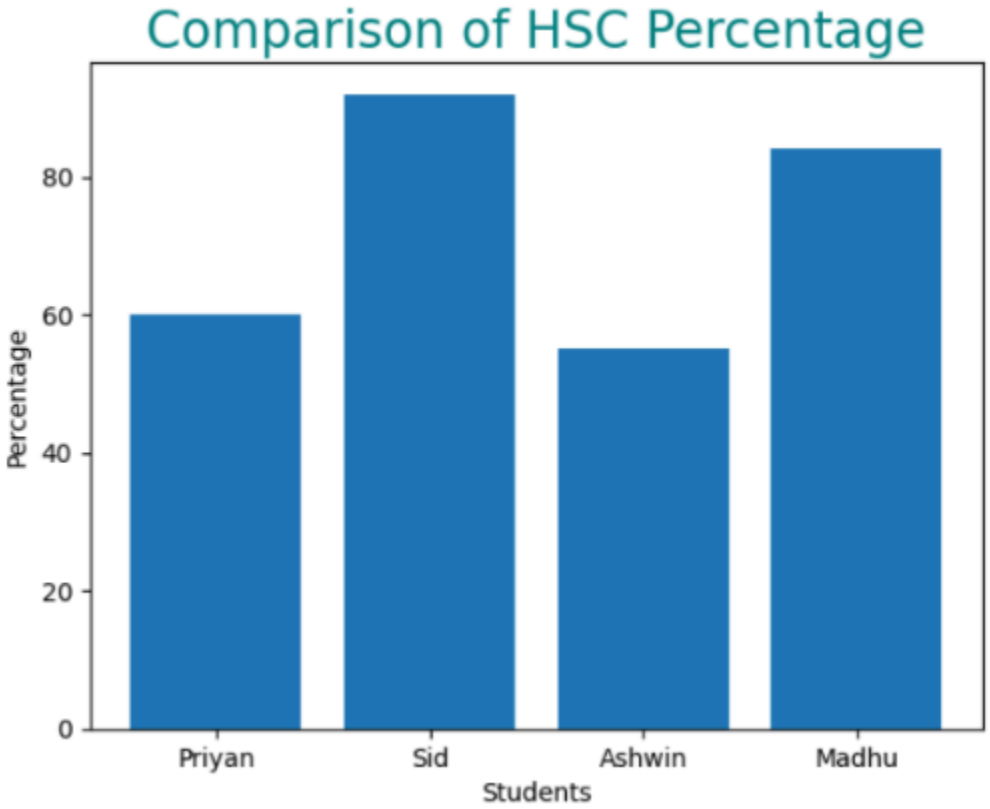
DESCRIPTION:

Take any sample data either through csv file or data fetched directly through code.

CODE:

```
import matplotlib.pyplot as hscmark
import numpy as np
Names = ['Priyan', 'Sid', 'Ashwin', 'Madhu']
xaxis = np.arange(len(Names))
per_hsc = [60,92,55,84]
hscmark.bar(Names, per_hsc)
hscmark.xticks(xaxis, Names)
hscmark.xlabel("Students")
hscmark.ylabel("Percentage")
hscmark.title("Comparison of HSC Percentage", fontsize=20, color="teal")
hscmark.show()
```

OUTPUT:



RESULT:

The data visualization of HSC percentage was successfully performed using a bar chart.

3.c Data visualization using pie chart

AIM:

Conduct an experiment to show data visualization using pie chart

DESCRIPTION:

Take any sample data either through csv file or data fetched directly through code.

CODE:

```
import matplotlib.pyplot as election

labels = ['CANDIDATE 1', 'CANDIDATE 2', 'CANDIDATE 3']

Votes = [210,300,320]

colors = ['green', 'yellow', 'red']

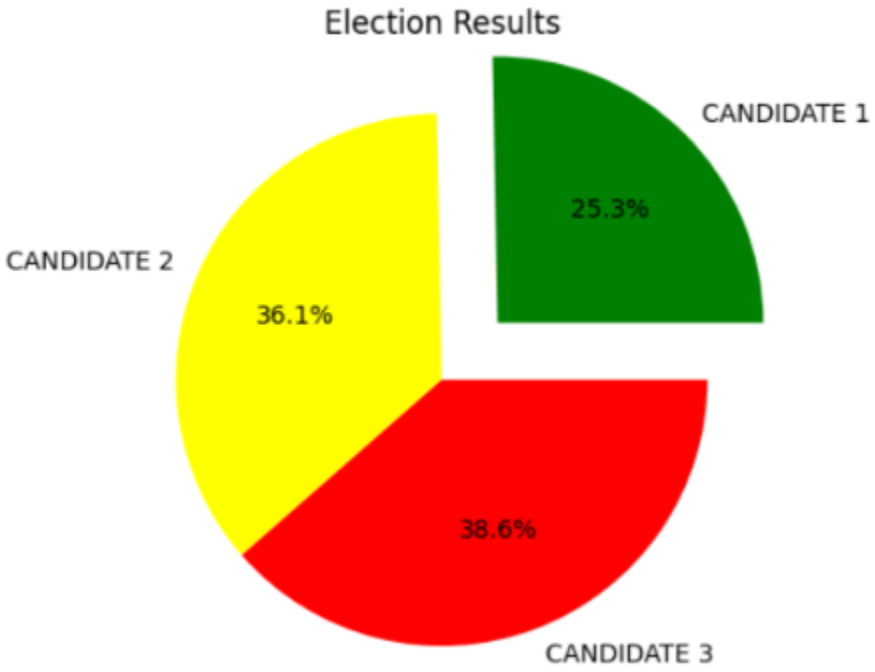
explode = (0.3, 0, 0, )

election.pie(Votes, labels=labels, colors=colors, explode=explode, autopct='%0.1f%%')

election.title('Election Results')

election.show()
```

OUTPUT:



RESULT:

The data visualization of election results was successfully performed using a pie chart.